DIFFERENTIATION

(1) ~~Sum Rule / Difference Rule~~

$$\cancel{\frac{d}{dx}\{f(x) \pm g(x)\} = \frac{df(x)}{dx} \pm \frac{dg(x)}{dx}}$$

Derivative of function

Suppose f is a real valued function and a is a point in its domain of definition. The derivative of f at a is defined by

$$\lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

Provided this limit exists. Derivative of f(x) at a is denoted by f'(a).

Example: Find the derivative at $x = 2$ of the function $f(x) = 3x$.

Solution: We have

$$f'(2) = \lim_{h \to 0} \frac{f(2+h) - f(2)}{h}$$

$$= \lim_{h \to 0} \frac{3(2+h) - 3(2)}{h} = \lim_{h \to 0} \frac{6 + 3h - 6}{h}$$

$$= \lim_{h \to 0} \frac{3h}{h} = \lim_{h \to 0} 3 = 3$$

Derivative of fn at $x = 2$ is 3.

Q2. Find derivative of the function $f(x) = 2x^2 + 3x - 5$ at $x = -1$. Also prove that $f'(0) + 3f'(-1) = 0$.

Solution. We first find derivative of $f(x)$ at $x = -1$.

We have

$$f'(-1) = \lim_{h \to 0} \frac{f(-1+h) - f(-1)}{h}$$

$$= \lim_{h \to 0} \frac{[2(-1+h)^2 + 3.(-1+h) - 5] - [2(-1)^2 + 3(-1) - 5]}{h}$$

$$= \lim_{h \to 0} \frac{[2(1+h^2-2h) - \cancel{3} + 3h - \cancel{5} - 2 + \cancel{3} + \cancel{5}]}{h}$$

$$= \lim_{h \to 0} \frac{\cancel{2} + 2h^2 - 4h + 3h - \cancel{2}}{h}$$

$$= \lim_{h \to 0} \frac{2h^2 - h}{h} = \lim_{h \to 0} \frac{\cancel{h}(2h-1)}{\cancel{h}}$$

$$= \lim_{h \to 0} 2h - 1 = 2(0) - 1 = -1.$$

Q3. Find derivative of $\sin x$ at $x = 0$

Let $f(x) = \sin x$. Then

$$f'(0) = \lim_{h \to 0} \frac{f(0+h) - f(0)}{h}$$

$$= \lim_{h \to 0} \frac{\sin(0+h) - \sin 0}{h} = \lim_{h \to 0} \frac{\sin h}{h}$$

$$= 1.$$

(3) Find the derivative of $f(x) = 3$ at $x=0$ and at $x=3$.

$$f'(0) = \lim_{h\to 0} \frac{f(0+h)-f(0)}{h} = \lim_{h\to 0} \frac{3-3}{h} = \lim_{h\to 0} \frac{0}{h} = 0.$$

$$\text{III}^{ly}\ f'(3) = \lim_{h\to 0} \frac{f(3+h)-f(3)}{h} = \lim_{h\to 0} \frac{3-3}{h}$$

$$= \lim_{h\to 0} \frac{0}{h} = 0.$$

Algebra of derivative of function.

i) $\frac{d}{dx}[f(x)+g(x)] = \frac{d}{dx}f(x) + \frac{d}{dx}g(x)$. (SUM)

(ii) $\frac{d}{dx}[f(x)-g(x)] = \frac{d}{dx}f(x) - \frac{d}{dx}g(x)$ (DIFFERENCE)

(iii) $\frac{d}{dx}[f(x).g(x)] = \frac{d}{dx}f(x).g(x) + f(x)\frac{d}{dx}g(x)$.

iv) $\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{\frac{d}{dx}f(x)\,g(x) - f(x)\frac{d}{dx}g(x)}{(g(x))^2}$.

~~v) Leibnitz rule:~~

~~$u = f(x)$ and $v = g(x)$~~

~~$(uv)' = u'v + uv'$.~~

~~$\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$.~~

Derivative of polynomials

Let $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$ be a polynomial function, where a_i's are all real numbers and $a_n \neq 0$. Then derivative function is given by

$$\frac{df}{dx} = n a_n x^{n-1} + (n-1) a_{n-1} x^{n-2} + \dots + 2a_2 x + a_1 .$$

Example: Compute the derivative of $6x^{100} - x^{55} + x$.

$$\therefore \frac{d(6x^{100} - x^{55} + x)}{dx} = 600x^{99} - 55x^{54} + 1$$

Note: If $f(x) = x^n$, then

$$\boxed{\frac{df(x)}{dx} = \frac{dx^n}{dx} = n x^{n-1}}$$

UNIT - III

DIFFERENTIATION

Some standard Differentiation

1) Differentiation of algebraic function (Polynomial and rational functions)

a) $\frac{d}{dx} x^n = n x^{n-1}, \quad x \in \mathbb{R}, n \in \mathbb{R}, x > 0$

b) $\frac{d}{dx}(\sqrt{x}) = \frac{1}{2\sqrt{x}}$

c) $\frac{d}{dx}\left(\frac{1}{x^n}\right) = -\frac{n}{x^{n+1}}$

2) Differentiation of trigonometric functions

a) $\frac{d}{dx} \sin x = \cos x$ b) $\frac{d}{dx} \cos x = -\sin x$

c) $\frac{d}{dx} \tan x = \sec^2 x$ d) $\frac{d}{dx} \sec x = \sec x \tan$

e) $\frac{d}{dx} \operatorname{cosec} x = -\operatorname{cosec} x \cot x$ f) $\frac{d}{dx} \cot x = -\operatorname{cosec}^2 x$

3) Differentiation of logarithmic and exponential function

a) $\frac{d}{dx}\log x = \frac{1}{x}$, for $x > 0$

b) $\frac{d}{dx} e^x = e^x$

c) $\frac{d}{dx} a^x = a^x \log a$, for $a > 0$

d) $\frac{d}{dx}\log_a x = \frac{1}{x \log a}$, for $x > 0$, $a > 0$ & $a \neq 1$.

4) Differentiation of Inverse trignometric function

a) $\frac{d}{dx}\sin^{-1} x = \frac{1}{\sqrt{1-x^2}}$, for $-1 < x < 1$

b) $\frac{d}{dx}\cos^{-1} x = \frac{-1}{\sqrt{1-x^2}}$, for $-1 < x < 1$

c) $\frac{d}{dx}\sec^{-1} x = \frac{1}{|x|\sqrt{x^2-1}}$, for $|x| > 1$

d) $\frac{d}{dx} \operatorname{cosec}^{-1} x = \frac{-1}{|x|\sqrt{x^2-1}}$ for $|x| > 1$.

e) $\frac{d}{dx} \tan^{-1} x = \frac{1}{1+x^2}$, for $x \in \mathbb{R}$

f) $\frac{d}{dx} \cot^{-1} x = \frac{-1}{1+x^2}$, for $x \in \mathbb{R}$

Chain Rule :

If y is a function of u and u is a function of x, then derivative of y with respect to x is

$$\boxed{\frac{dy(u)}{dx} = \frac{dy}{du} \times \frac{du}{dx}}$$

Example: Find the derivatives of the function $f(x) = \sin(\underbrace{2x^2 - 6x}_{\text{this is another function say } u(x)})$

Solution : Given $f(x) = \sin(2x^2 - 6x)$

$$\therefore \frac{df(x)}{dx} = \frac{d\sin(2x^2-6x)}{d(2x^2-6x)} \times \frac{d(2x^2-6x)}{dx}$$

$$= \cos(2x^2-6x) \times \left[\frac{d2x^2}{dx} - \frac{d6x}{dx}\right]$$

$$= \cos(2x^2-6x)\left[4x-6\right]$$

$$= \cos(2x^2-6x)(4x-6)$$

Example: Find the derivative of the function given by $f(x) = \sin(e^{x^3})$

Solution : Using chain rule

$$\frac{df(x)}{dx} = \frac{d\sin(e^{x^3})}{d(x^3)} \times \frac{d(x^3)}{dx}$$

$$= \cos(e^{x^3}) \times 3x^2$$

$$\frac{df(x)}{dx} = 3x^2\cos(e^{x^3})$$

Practice Questions on chain rule:

Find the derivative of the function

a) $y = \cos^2(x^4)$

b) $y = \sin^4 x + \sin x^4$

c) $y = 2\log[\log(\log \sec x)]$

Derivatives of Implicit functions

An implicit function is a function, written in terms of both dependent and independent variables, like

$y - 3x^2 + 2x + 5 = 0$.

Note: Implicit function can be differentiated using chain rule, or product rule or quotient rule.

Example: Find $\frac{dy}{dx}$ if $y = 5x^2 - 9y$.

Solution: $y = 5x^2 - 9y$

$$\Rightarrow \frac{dy}{dx} = \frac{d(5x^2)}{dx} - \frac{d(9y)}{dy} \times \frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx} = 10x - 9\frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx} + 9\frac{dy}{dx} = 10x$$

$$\frac{dy}{dx}\left[1+9\right] = 10x$$

$$\frac{dy}{dx} = \frac{10x}{10} = x.$$

$$\Rightarrow \boxed{\frac{dy}{dx} = x}$$

Example: Find $\frac{dy}{dx}$ if $y = 5x^2 - 9e^y$

Solution: $y = 5x^2 - 9e^y$

$$\frac{dy}{dx} = \frac{d(5x^2)}{dx} - \frac{d(9e^y)}{dy} \times \frac{dy}{dx}.$$

$$\Rightarrow \frac{dy}{dx} = 10x - 9e^y\frac{dy}{dx} \Rightarrow \frac{dy}{dx} + 9e^y\frac{dy}{dx} = 10x$$

Logarithmic function

Let a given function be

$$y = f(x) = [u(x)]^{\{v(x)\}}$$

Steps to differentiate these kind of function

Step 1: Take logarithm (to base e) to the both side of given function, we get

$$\log y = \log (u(x))^{\{v(x)\}}$$
$$= v(x) \log(u(x))$$

Step 2: Now using chain rule and product rule, differentiate the function

$$\frac{d \log y}{dy} \times \frac{dy}{dx} = \frac{d(v(x)\log(u(x)))}{dx}$$

$$\Rightarrow \frac{1}{y}\frac{dy}{dx} = v(x)\frac{d\log(u(x))}{dx} + \log(u(x))\frac{dv(x)}{dx}$$

$$\Rightarrow \frac{1}{y}\frac{dy}{dx} = \frac{v(x)}{u(x)}\frac{du(x)}{dx} + \log(u(x))\frac{dv}{dx}$$

$$\Rightarrow \boxed{\frac{dy}{dx} = y\left[\frac{v(x)}{u(x)}\frac{du(x)}{dx} + \frac{dv(x)}{dx}\log(u(x))\right]}$$

Example: Find the value of $\frac{dy}{dx}$ if,

$$y = e^{x^4}$$

Solution: $y = e^{x^4}$

Taking logarithm of both the sides we get,

$$\log y = \log e^{x^4}$$
$$\log y = x^4 \log e$$
$$\log y = x^4 \qquad [\because \log_e e = 1]$$

Taking logarithm of both the sides we get,

$$\log y = \log e^{x^4}$$

$$\log y = x^4 \log e$$

$$\log y = x^4 \qquad [\because \log_e e = 1]$$

Now, differentiating both the sides w.r.t we get,

$$\frac{d \log y}{dy} \times \frac{dy}{dx} = \frac{d x^4}{dx}$$

$$\frac{1}{y}\frac{dy}{dx} = 4x^3$$

$$\Rightarrow \boxed{\frac{dy}{dx} = 4x^3 y}$$

Example : Find $\frac{dy}{dx}$ if $y = 2x^{\cos x}$

Solution : Given $y = 2x^{\cos x}$

$$\log y = \log(2x^{\cos x})$$

$$= \log 2 + \log x^{\cos x} \qquad [\because \log(ab) = \log a + \log b]$$

$$\log y = \log 2 + \cos x \log x \qquad [\because \log x^n = n \log x]$$

Differentiating both side

$$\frac{d \log y}{dy} \times \frac{dy}{dx} = \frac{d \log 2}{dx} + \frac{d(\cos x \log x)}{dx}$$

$$\frac{1}{y}\frac{dy}{dx} = 0 + \cos x \frac{d \log x}{dx} + \log x \frac{d \cos x}{dx}$$

$$\frac{1}{y}\frac{dy}{dx} = \frac{\cos x}{x} + \log x(-\sin x)$$

$$\Rightarrow \boxed{\frac{dy}{dx} = y\left[\frac{\cos x}{x} - \sin x \log x\right]}$$

Practice Question:

a) Find derivative

$$y = x^x$$

b) $$y = \sqrt{\frac{(x-3)(x^2+4)}{(3x^2+4x+5)}}$$

c) $$y = a^x$$

Differentiation by substitution

(7) Some suitable substitutions

S. N.	Function	Substitution	S. N.	Function	Substitution
(i)	$\sqrt{a^2 - x^2}$	$x = a\sin\theta$ or $a\cos\theta$	(ii)	$\sqrt{x^2 + a^2}$	$x = a\tan\theta$ or $a\cot\theta$
(iii)	$\sqrt{x^2 - a^2}$	$x = a\sec\theta$ or $a\,\text{cosec}\,\theta$	(iv)	$\sqrt{\frac{a-x}{a+x}}$	$x = a\cos 2\theta$
(v)	$\sqrt{\frac{a^2 - x^2}{a^2 + x^2}}$	$x^2 = a^2\cos 2\theta$	(vi)	$\sqrt{ax - x^2}$	$x = a\sin^2\theta$
(vii)	$\sqrt{\frac{x}{a+x}}$	$x = a\tan^2\theta$	(viii)	$\sqrt{\frac{x}{a-x}}$	$x = a\sin^2\theta$
(ix)	$\sqrt{(x-a)(x-b)}$	$x = a\sec^2\theta - b\tan^2\theta$	(x)	$\sqrt{(x-a)(b-x)}$	$x = a\cos^2\theta + b\sin^2\theta$

For example: Find the derivative of

$$\tan^{-1}\left[\frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}}\right]$$ with respect to x.

Solution: Put $x^2 = \cos 2\theta$

$$y = \tan^{-1}\left[\frac{\sqrt{1+\cos 2\theta} + \sqrt{1-\cos 2\theta}}{\sqrt{1+\cos 2\theta} - \sqrt{1-\cos 2\theta}}\right]$$

∵ Use the identity

$$1 + \cos 2\theta = 2\cos^2\theta$$
$$1 - \cos 2\theta = 2\sin^2\theta$$

$$y = \tan^{-1}\left[\frac{\sqrt{2\cos^2\theta} + \sqrt{2\sin^2\theta}}{\sqrt{2\cos^2\theta} - \sqrt{2\sin^2\theta}}\right]$$

$$= \tan^{-1}\left[\cos\theta\sqrt{2} + \sin\theta\sqrt{2}\right.$$

$$y = \tan^{-1}\left[\frac{\sqrt{2\cos^2\theta} + \sqrt{2\sin^2\theta}}{\sqrt{2\cos^2\theta} - \sqrt{2\sin^2\theta}}\right]$$

$$= \tan^{-1}\left[\frac{\cos\theta\sqrt{2} + \sin\theta\sqrt{2}}{\cos\theta\sqrt{2} - \sin\theta\sqrt{2}}\right]$$

$$= \tan^{-1}\left[\frac{\cancel{\sqrt{2}}}{\cancel{\sqrt{2}}}\left[\frac{\cos\theta + \sin\theta}{\cos\theta - \sin\theta}\right]\right]$$

$$= \tan^{-1}\left[\frac{\cos\theta + \sin\theta}{\cos\theta - \sin\theta}\right]$$

Now dividing $\cos\theta$ in nume
and denominator

$$\Rightarrow y = \tan^{-1}\left[\frac{\cos\theta + \sin\theta}{\cos\theta - \sin\theta} \times \frac{\cos\theta}{\cos\theta}\right]$$

$$= \tan^{-1}\left[\frac{\frac{\cos\theta + \sin\theta}{\cos\theta}}{\frac{\cos\theta - \sin\theta}{\cos\theta}}\right]$$

$$\Rightarrow y = \tan^{-1}\left[\frac{\frac{\cos\theta}{\cos\theta} + \frac{\sin\theta}{\cos\theta}}{\frac{\cos\theta}{\cos\theta} - \frac{\sin\theta}{\cos\theta}}\right]$$

$$= \tan^{-1}\left[\frac{1 + \tan\theta}{1 - \tan\theta}\right] \quad \left[\because \tan = \frac{\sin\theta}{\cos\theta}\right]$$

$$\Rightarrow y = \tan^{-1}\left(\tan\left(\frac{\pi}{4} + \theta\right)\right)$$

$$\Rightarrow y = \frac{\pi}{4} + \theta$$

$$\Rightarrow y = \frac{\pi}{4} + \frac{1}{2}\cos^{-1}x^2$$

On differentiating with respect to x we get

$$\frac{dy}{dx} = \frac{d}{dx}\left(\frac{\pi}{4}\right) + \frac{1}{2}\frac{d}{dx}\left(\cos^{-1}x^2\right)$$

$$= \tan^{-1}\left[\frac{1+\tan\theta}{1-\tan\theta}\right] \quad \left[\because \tan\theta = \frac{\sin\theta}{\cos\theta}\right]$$

$$\Rightarrow y = \tan^{-1}\left(\tan\left(\frac{\pi}{4}+\theta\right)\right)$$

$$\Rightarrow y = \frac{\pi}{4}+\theta$$

$$\Rightarrow y = \frac{\pi}{4}+\frac{1}{2}\cos^{-1}x^2$$

On differentiating with respect to x we get

$$\frac{dy}{dx} = \frac{d}{dx}\left(\frac{\pi}{4}\right) + \frac{1}{2}\frac{d}{dx}\left(\cos^{-1}x^2\right)$$

$$\Rightarrow \frac{dy}{dx} = 0 + \frac{1}{2}\frac{-1}{\sqrt{1-x^2}}\frac{d}{dx}(x^2)$$

$$\frac{dy}{dx} = -\frac{1}{2}\frac{2x}{\sqrt{1-x^2}} = \frac{-x}{\sqrt{1-x^2}}$$

Practice Question on substitution method.

Find the derivative

a) $y = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$

b) $y = \tan^{-1}\left(\frac{6x}{1-9x^2}\right.$